Symbolic Music Generation with REMI Transformers

DeepMIR 2026 Fall — Homework 2

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Code, README: https://github.com/chijenpeng/26-fall-DeepMIR-HW2

Checkpoints, generated results:
https://github.com/chijenpeng/26-fall-DeepMIR-HW2/releases/tag/submission-v1

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GPU Setting

GPU setting

Basic GPT-2	NVIDIA RTX A4500 (20 GB), Intel i9-14900K, 125 GB
Bonus T5, no-chord	NVIDIA RTX 4070 SUPER (12 GB), Intel i7-8700, 60 GB
OS	Ubuntu 26.04
Software	PyTorch 2.8, Transformers 4.57.1, pretty_midi 0.2.11, FluidSynth

Basic Task A: Unconditional Generation

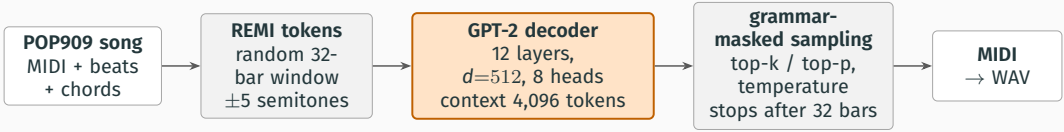
Basic A (1): token representation — REMI

- REMI (Huang & Yang, 2020): `Bar`, `Pos` (16th note), `Chord`, `Tempo`, `Pitch`, `Vel`, `Dur`
- Bars follow the downbeats annotated in `beat_midi.txt`; a `Beat_n` token gives each bar's length
- MELODY, BRIDGE and PIANO merged into one stream; 430 tokens in the vocabulary
- 32 bars $\approx$ 2,431 tokens (median)

Example (song 001, bar 3)

```
Bar Beat_4 Tempo_90 Pos_0 Chord_A#:min Pitch_46 Vel_21 Dur_7 Pitch_75 Vel_29
Dur_2 Pos_1 Pitch_53 Vel_18 Dur_5 Pos_2 …
```

Basic A (2): model architecture



Model	`GPT2LMHeadModel`
Layers / width / heads	12 / 512 / 8
Context	4,096 tokens
Vocabulary	430
Parameters	**40.1 M**, random init

- Decoder-only language model, next-token cross-entropy
- 4,096 positions cover a whole 32-bar window (p99 3,900 tokens)

Basic A (3): implementation details

Data and augmentation

- All 896 training songs, random 32-bar windows
- Random transposition by $-5 \cdots +5$ semitones, pitches and chord roots together

Optimizer	AdamW, weight decay 0.01
Learning rate	3×10^{-4}
Schedule	500 warm-up, cosine to 10%
Batch	4 $\times$ 4 accumulation = 16
Steps	6,000
Precision	bf16, grad-norm clip 1.0
Training time	$\approx$2.6 h to step 5,000

Basic A (4): checkpoint selection

Step	With chords	No chords
3k	1.358	1.354
4k	1.323	1.319
5k	**1.308**	**1.301**
6k	1.322	1.305

Held-out loss of the 6k-step run

- Held-out loss: token cross-entropy on the 3 Basic B prompt songs, which are not in the training set
- Lowest at step 5k for both models; it rises again at 6k

Final checkpoints: step 5,000

- With chords: `models/basic.pt`, used for Basic A and Basic B
- No chords (ablation): `models/basic-no-chord.pt`

Basic A (5): inference configurations

Setting	Sampling	Temperature
A	top-k, $K = 20$	0.9
B	top-p, $P = 0.9$	1.0
C	top-k, $K = 50$	1.2

- Checkpoint: `models/basic.pt` (step 5,000)
- 20 pieces $\times$ 32 bars per setting, piece *i* uses seed $2026 + i$
- Ablation: the model trained **without chord tokens**, sampled with setting A

Grammar mask

- Sampling is restricted to tokens that can come next in REMI, e.g. `Pitch` $\rightarrow$ `Vel` $\rightarrow$ `Dur`
- `Pos` strictly increases inside a bar and stays below its beat count
- Generation stops after exactly 32 bars

Basic A (6): objective metrics

Model	Repr.	w/ Chord	Loss	Top-K / Top-P	Temp.	H1	H4	GS
GPT-2	REMI	O	1.043	K = 20	0.9	2.197	**2.604**	0.909
GPT-2	REMI	O	1.043	P = 0.9	1.0	2.212	2.557	0.947
GPT-2	REMI	O	1.043	K = 50	1.2	2.326	2.801	**0.849**
GPT-2	REMI	X	1.051	K = 20	0.9	2.146	2.550	0.943
Real data (896 songs)						2.123	2.626	0.877

- **A**: closest H4, smallest $|\Delta \mathrm{H4}| + |\Delta \mathrm{GS}|$ (0.054; B 0.139, C 0.203) $\rightarrow$ submitted listening samples
- Higher temperature (**C**) gives more erratic pitch (H1/H4 above real data) and less regular rhythm (GS below)
- Top-p (**B**) gives more repetitive rhythm (GS 0.947 $>$ 0.877)
- Without chord tokens (setting A): H1 closer to the real data, H4 and GS farther

Loss: training cross-entropy of the step-5k checkpoint (mean of steps 4.8k–5k).

20 pieces $\times$ 32 bars per row; H1, H4, GS from the TA `eval_basic.py` (MusDr); closer to the real data is better.

Basic Task B: Prompt Continuation

Basic B: continuation of the three prompts

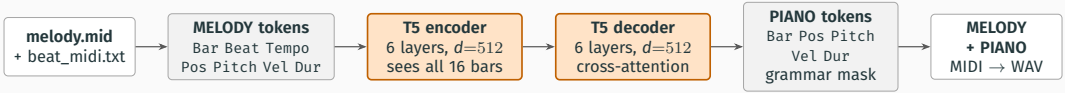
Prompt	Setting	Notes	Tempo (prompt → cont.)	Best
001	A: K = 20, T 0.9	433	63 → 63	✓
	B: P = 0.9, T 1.0	387	63 → 63	
	C: K = 50, T 1.2	559	63 → 51–63	
002	A	443	108 → 108	✓
	B	445	108 → 108	
	C	458	108 → 108	
003	A	851	69 → 69	✓
	B	669	69 → 69	
	C	724	69 → 69	

Model: the Basic A GPT-2 checkpoint, REMI with chords. By listening, all three settings sound equally good for every prompt; we name setting A the best for all three, the same setting submitted for Basic A.

8 prompt bars kept verbatim + 24 generated bars.

Bonus: Accompaniment Generation

Bonus (1): model and representation — Option B, seq2seq



Model	`T5ForConditionalGeneration`
Layers	6 encoder + 6 decoder
Width / heads / FFN	512 / 8 / 2,048
Max length	1,024 melody / 2,048 piano
Parameters	**44.3 M**, random init

- Same REMI vocabulary as Basic, but no chord tokens: the harmony must come from the melody alone
- Encoder: MELODY track with the bar headers, so it knows meter and tempo
- Decoder: PIANO track; bar headers are omitted because the piano is placed on the prompt's own beat grid
- Inference reads only `melody.mid` and `beat_midi.txt`; the melody is kept verbatim
- 16 bars: median 316 melody / 758 piano tokens

Bonus (2): implementation details and checkpoint

Data and augmentation

- All 896 training songs; random 16-bar windows, MELODY $\rightarrow$ PIANO pairs
- Random transposition by $-5 \cdots +5$ semitones (both tracks)

Optimizer	AdamW, weight decay 0.01
Learning rate	3×10^{-4}
Schedule	500 warm-up, cosine over 80k
Batch	2×8 accumulation = 16
Precision	bf16, grad-norm clip 1.0
Training time	$\approx$1.7 h to step 12,000

Checkpoint selection

Step	Held-out loss
4k	1.156
12k	**1.076**
20k	1.258
28k	1.490

Final checkpoint: step 12,000

- `models/bonus.pt`, lowest held-out loss

Held-out loss: decoder cross-entropy on the 10 Bonus prompt songs, which are not in the training set.

Bonus (3): inference configurations and objective metrics

Setting	Sampling, temperature	CA	OA_P	OA_V	OA_D	OA_{IOI}
A (submitted)	top-k 20, T 0.9	0.150	0.844	**0.597**	0.695	**0.609**
B	top-p 0.9, T 1.0	0.156	0.850	0.556	0.674	0.561
C	top-k 50, T 1.2	0.106	0.843	0.541	0.681	0.576
D	top-k 5, T 0.7	0.150	**0.851**	0.567	0.729	0.578
E	top-k 3, T 0.5	**0.163**	0.849	0.568	**0.736**	0.572
Ground-truth piano (through our tokenizer)		0.750	1.00	1.00	0.96	1.00

- **A** is submitted: best OA on velocity and IOI, middle on the rest; lower temperature (D, E) helps duration and CA only slightly
- CA is low for every setting. The ground-truth piano passed through our tokenizer scores 0.750, so the pipeline is aligned; the model has not learned a reliable melody $\rightarrow$ harmony mapping

10 prompts $\times$ 16 bars, TA `eval_bonus.py`, averaged over bars; seeds 2026–2035.

Disclosure

Use of AI assistance

Tools: Claude Code with the Claude Opus 5.5 model (Anthropic); OpenAI Codex for an earlier version.

Done with AI assistance

- Code implementation: tokenizer, training, generation, evaluation
- Running the experiments on the servers and keeping the experiment log
- Preparing these slides

My own

- The decisions: which experiments to run, how to read them, which models and settings to submit
- Listening to the generated pieces and choosing the Basic B continuations

Every number in this report comes from running the code in the repository, and I reviewed the results and the text.

References

References

- **POP909**: Z. Wang et al., “POP909: A Pop-song Dataset for Music Arrangement Generation,” ISMIR 2020
- **REMI**: Y.-S. Huang, Y.-H. Yang, “Pop Music Transformer,” ACM MM 2020
- **MusDr** metrics: S.-L. Wu, Y.-H. Yang, “The Jazz Transformer on the Front Line,” ISMIR 2020
- **Bonus metrics**: Y. Ren et al., “PopMAG: Pop Music Accompaniment Generation,” ACM MM 2020
- **GPT-2**: A. Radford et al., 2019. **T5**: C. Raffel et al., JMLR 2020
- Hugging Face Transformers: T. Wolf et al., EMNLP 2020. PyTorch: A. Paszke et al., NeurIPS 2019
- TA scripts (`tokenization.py`, `eval_basic.py`, `eval_bonus.py`); TimGM6mb SoundFont (GPL-2)